

# *Operations Research & Applications programming*

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## **Ch. 2**

### **Modeling with linear programming**

# *Introduction*

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- *This chapter*, starts with the modeling and graphical solution of a two variable problem which, though highly simplified, provides a concrete understanding of the basic concepts of LP and lays the foundation for the development of the general *simplex* algorithm in Chapter 3
- On the computational side, TORA, a totally menu-driven and self-documenting tutorial program, is designed to help you understand the basics of LP through interactive feedback

## ***Tow - Variable LP Model***

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- ❑ This section deals with the graphical solution of a two-variable LP. Though two-variable problems hardly exist in practice.
- ❑ The LP model, as in any OR model, has three basic components.
  1. Decision variables that we seek to determine.
  2. Objective (goal) that we need to optimize (maximize or minimize).
  3. Constraints that the solution must satisfy.

## ***Tow - Variable LP Model***

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- ***Example 2.1:*** Wild West produces two types of cowboy hats. Type I hat requires twice as much labor as a Type II. If all the available labor time is dedicated to Type II alone, the company can produce a total of 400 Type II hats a day. The respective market hats are 150 and 200 hats per day limits for the two types of. The profit is \$8 per Type I hat and \$5 per Type II hat.

Formulate the problem as a LP so as to maximize the profit.

## ***Tow - Variable LP Model***

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❑ **Solution:** Assume that Wild West produces  $x_1$  Type I hats and  $x_2$  Type II hats per day. **Per day Profit got**  $= 8 x_1 + 5 x_2$

❑ Assume the time spent in producing one type II hat is  $c$  m.

Labor Time spent is  $(2 x_1 + x_2) c$  minutes

❑ The **objective** is to find  $x_1, x_2$  so as to maximize the profit  $z = 8 x_1 + 5 x_2$  satisfying the **constraints:**

$$(2 x_1 + x_2) c \leq 400 c$$

$$x_1 \leq 150$$

$$x_2 \leq 200$$

$$x_1, x_2 \geq 0, \text{ integers}$$

## ***Tow - Variable LP Model***

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- *That is :* The **objective** is to find  $x_1, x_2$  so as to maximize the profit  $z = 8 x_1 + 5 x_2$  satisfying the **constraints**:

$$(2 x_1 + x_2 ) \leq 400 \quad (1)$$

$$x_1 \leq 150 \quad (2)$$

$$x_2 \leq 200 \quad (3)$$

$$x_1, x_2 \geq 0, \text{ integers.}$$

# *Graphical LP Solution*

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- The graphical procedure includes two steps:
  1. Determination of the feasible solution space.
  2. Determination of the optimum solution from among all the feasible points in the solution.
- The procedure uses two examples to show how maximization and minimization objective functions are handled.

## Graphical LP Solution

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- **Example 2.3 :** (The Reddy Miks Company)
- Reddy Mikks produces both interior and exterior paints from two raw materials,  $M1$  and  $M2$ . The following table provides the basic data of the problem:

|                         | Tons of raw material per ton of |                |                                   |
|-------------------------|---------------------------------|----------------|-----------------------------------|
|                         | interior paint                  | exterior paint | Maximum daily availability (tons) |
| Raw material, $M1$      | 6                               | 4              | 24                                |
| Raw material, $M2$      | 1                               | 2              | 6                                 |
| Profit per ton (\$1000) | 5                               | 4              |                                   |

## *Graphical LP Solution*

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- ❑ A market survey indicates that the daily demand for interior paint cannot exceed that for exterior paint by more than 1 ton. Also, the maximum daily demand for interior paint is 2 tons.
- ❑ Reddy Mikks wants to determine the optimum (best) product mix of interior and exterior paints that maximizes the total daily profit.
- ❑ ***Solution:*** For the Reddy Mikks problem, the variables of the model are defined as:  $x_1$ ( $x_2$ ) tons produced daily of interior (exterior) paint.

# Graphical LP Solution

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- Letting  $z$  represent the total daily profit (in thousands of dollars), the objective of the company is

$$\text{Maximize } z = 5x_1 + 4x_2$$

- Next, we construct the constraints that restrict raw material usage and product demand :

- Usage of raw material  $M1$  by both paints =  $6x_1 + 4x_2$  tons/day

- Usage of raw material  $M2$  by both paints =  $1x_1 + 2x_2$  tons/day

- Because the daily availabilities of raw materials  $M1$  and  $M2$  are limited to 24 and 6 tons, respectively, the associated restrictions are given as:

$$6x_1 + 4x_2 \leq 24$$

$$1x_1 + 2x_2 \leq 6$$

# Graphical LP Solution

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- ❑ The first demand restriction stipulates that the excess of the daily production of interior over exterior paint,  $x_1 - x_2$  should not exceed 1 ton, which translates to

$$x_1 - x_2 \leq 1 \quad (\text{Market limit})$$

- ❑ The second demand restriction stipulates that the maximum daily demand of interior paint is limited to 2 tons, which translates to  $x_1 \leq 2$  (Demand limit)

- ❑ The non-negativity restrictions,  $x_1 \geq 0$ ,  $x_2 \geq 0$ , account for this requirement.

# Graphical LP Solution

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□ That is : Maximize  $z = 5 x_1 + 4 x_2$

satisfying the constraints:

$$6 x_1 + 4 x_2 \leq 24 \quad (1)$$

$$x_1 + 2x_2 \leq 6 \quad (2)$$

$$x_1 - x_2 \leq 1 \quad (3)$$

$$x_1 \leq 2 \quad (4)$$

$$x_1, x_2 \geq 0$$

□ Any values of  $x_1$  and  $x_2$  that satisfy *all* five constraints constitute a *feasible solution*.

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# Graphical LP Solution

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## □ Step 1. Determination of the Feasible Solution Space:

- The non-negativity restrictions tell that the solution space is in the first quadrant. Then we replace each inequality constraint by an equality and then graph the resulting line (noting that two points will determine a line uniquely).
- Next we note that each constraint line divides the plane into two half-spaces and that on one half-space the constraint will be  $\leq$  and on the other it will be  $\geq$ .
- To determine the “correct” side we choose a reference point and see on which side it lies.

# *Graphical LP Solution*

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- ❑ (Normally  $(0,0)$  is chosen. If the constraint line passes through  $(0,0)$ , then we choose some other point.) Doing like this we would get the feasible solution space.
- ❑ *Step 2. Determination of the optimum Solution :*
- ❑ The determination of the optimal solution requires the direction in which the objective function will *increase (decrease)* in the case of a **maximization** (minimization) problem.

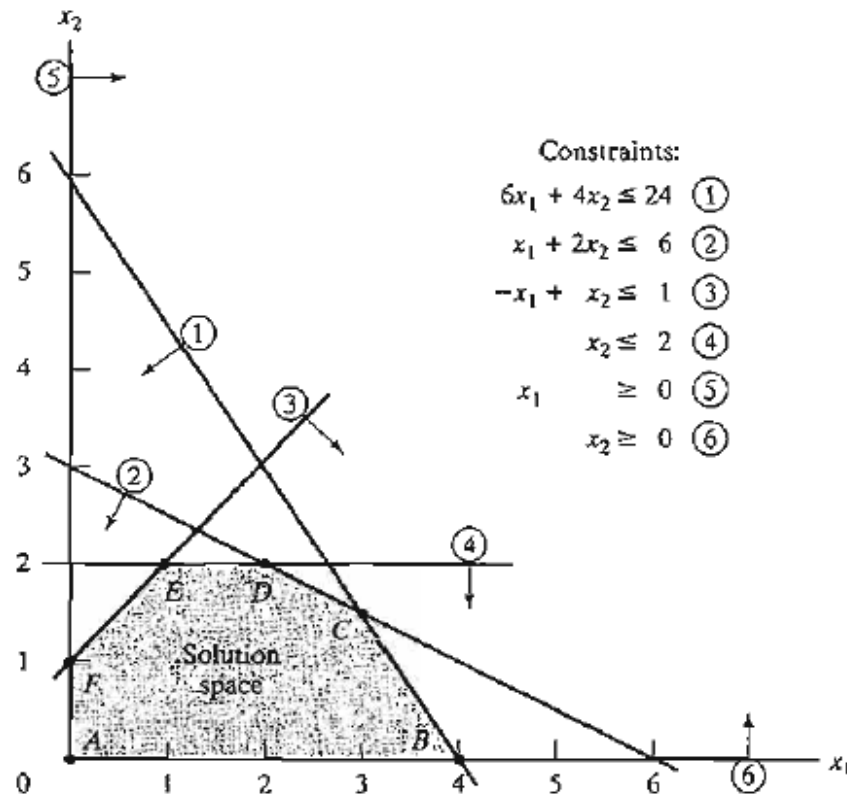
# *Graphical LP Solution*

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- ❑ We find this by assigning two increasing (decreasing) values for  $z$  and then drawing the graphs of the objective function for these two values.
- ❑ The *optimum solution* occurs at a point beyond which any further increase (decrease) of  $Z$  will make us leave the feasible space.

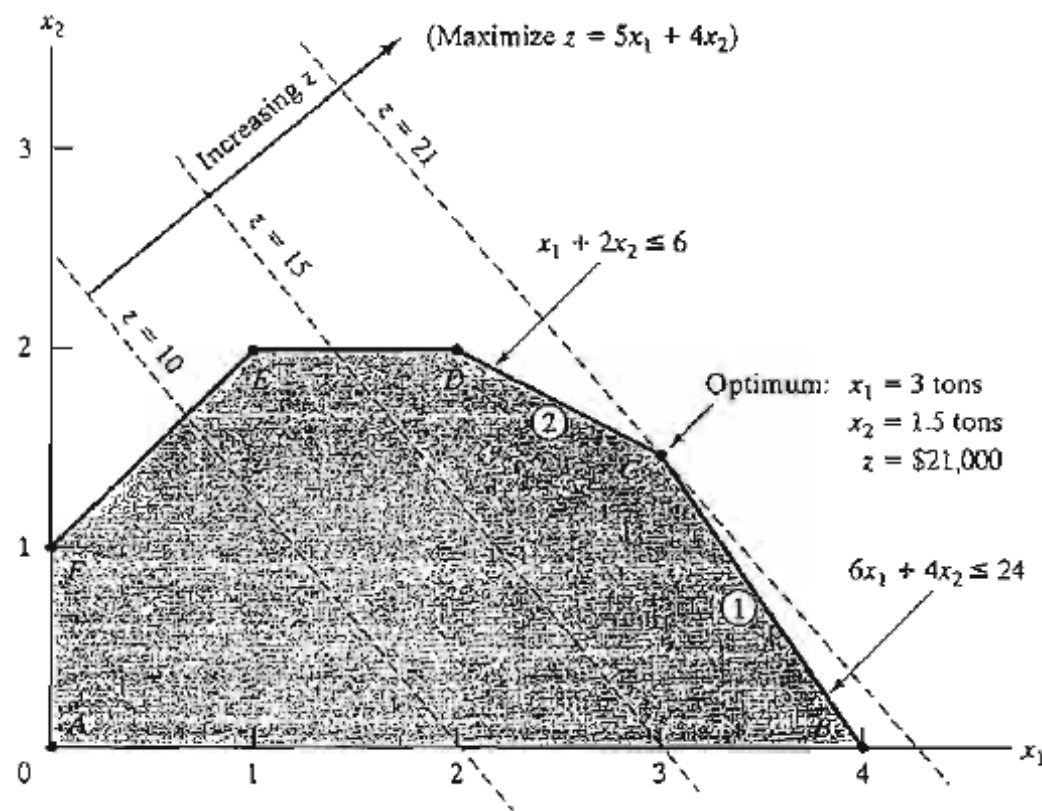
# Graphical LP Solution

□ **Step 1.** *Determination of the Feasible Solution Space:*



# Graphical LP Solution

□ **Step 2.** *Determination of the optimum Solution :*



## Graphical LP Solution Exam. 2.1

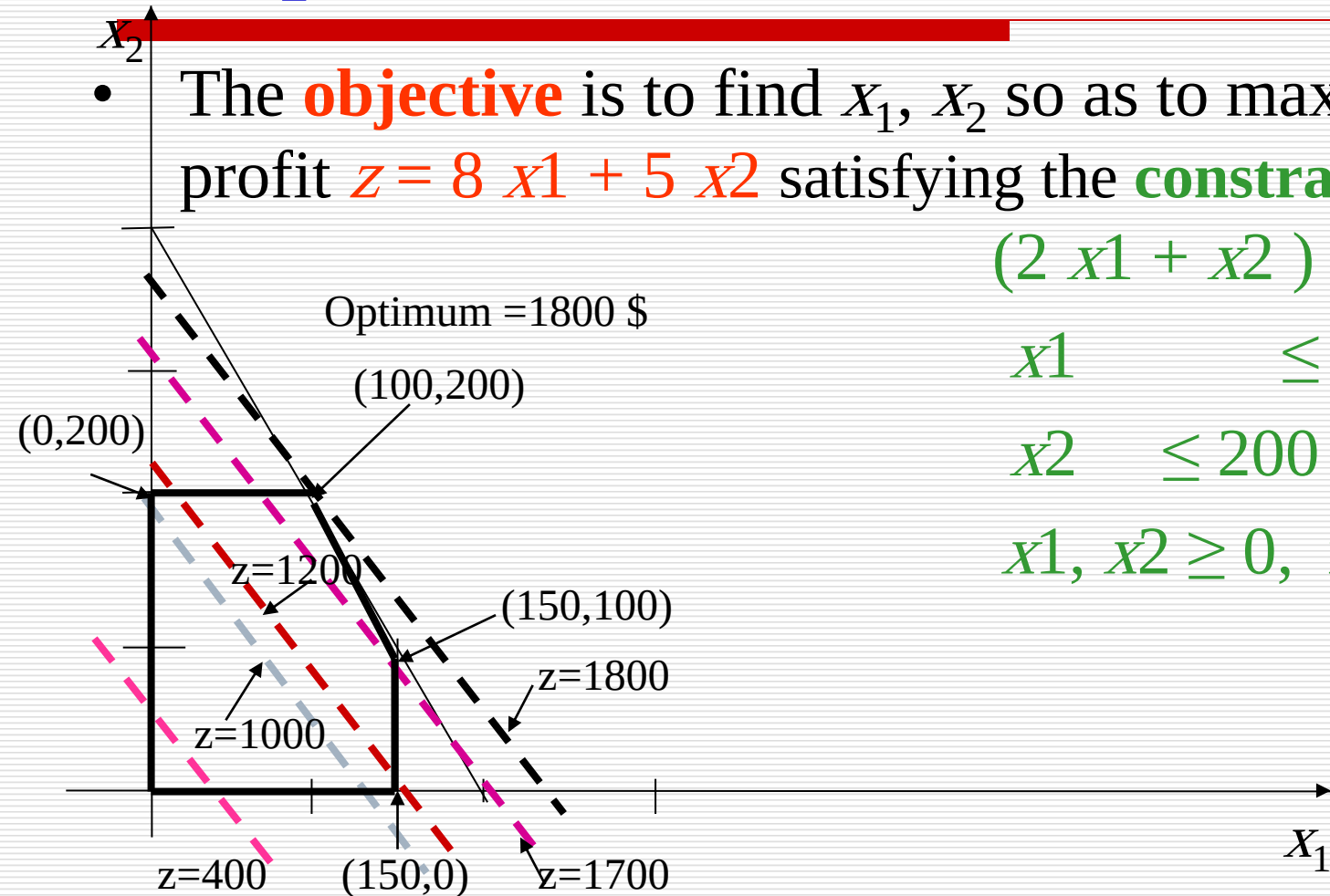
- The **objective** is to find  $x_1, x_2$  so as to maximize the profit  $z = 8x_1 + 5x_2$  satisfying the **constraints**:

$$(2x_1 + x_2) \leq 400$$

$$x_1 \leq 150$$

$$x_2 \leq 200$$

$$x_1, x_2 \geq 0, \text{ integers}$$



## *Graphical Solution of Feed Mix Problem*

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□ Minimize  $z = 2x_1 + 3x_2$

Subject to  $x_1 + 3x_2 \geq 15$

$2x_1 + 2x_2 \geq 20$

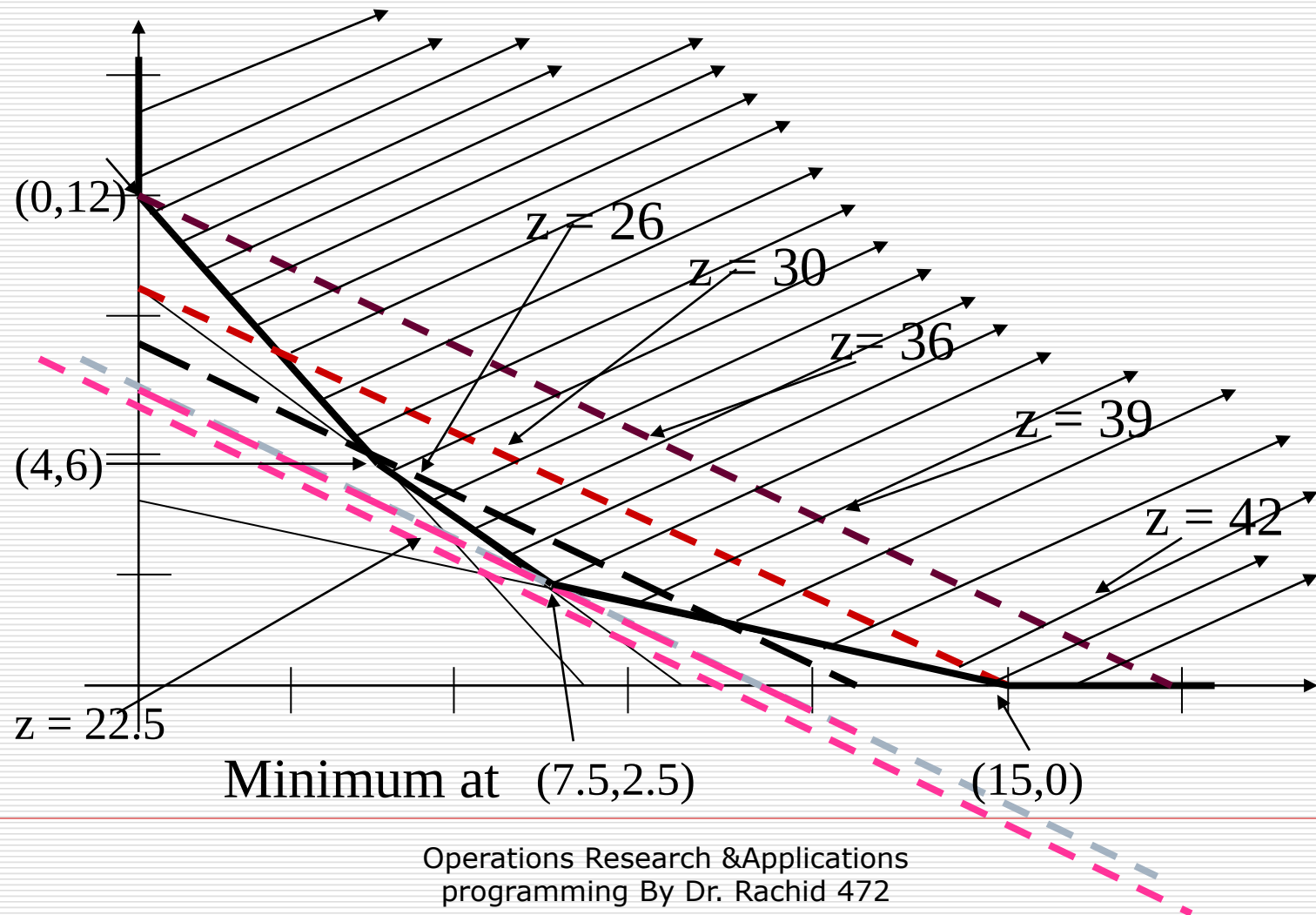
$3x_1 + 2x_2 \geq 24$

$x_1, x_2 \geq 0$

Optimum  $z = 22.5$  at  $(7.5, 2.5)$

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# *Graphical Solution of Feed Mix Problem*



## *Exceptional Cases*

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- Usually a LPP will have a unique optimal solution. But there are problems where there may be no solution, may have alternative optimum solutions and “unbounded” solutions. We graphically explain these cases in the following slides. We note that the (unique) optimum solution occurs at one of the “corners” of the set of all feasible points.

## *Alternative optimal solutions*

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□ Consider the LPP

$$\text{Maximum } z = 10x_1 + 5x_2$$

subject to the constraints:

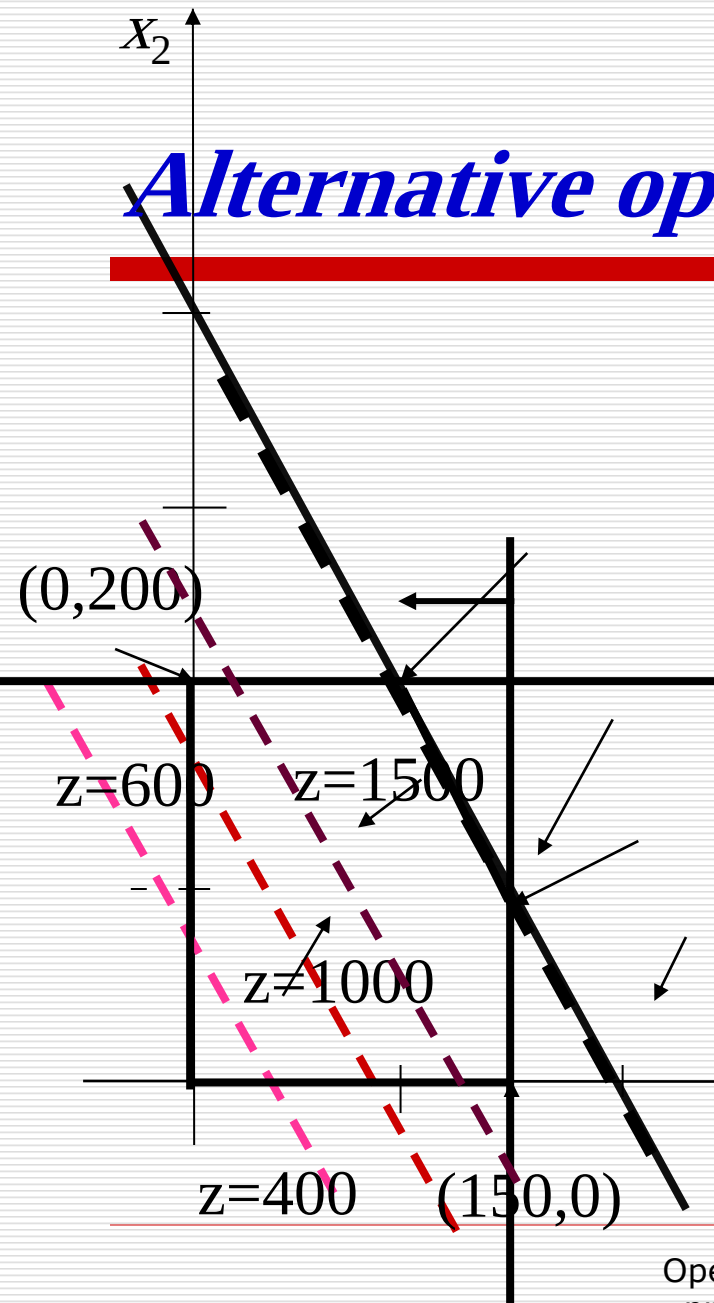
$$x_1 \leq 150$$

$$x_2 \leq 200$$

$$2x_1 + x_2 \leq 400$$

$$x_1, x_2 \geq 0$$

## *Alternative optimal solutions*



Thus we see that the objective function  $z$  is maximum at the corner  $(150, 100)$  and also has an alternative optimum solution at the corner  $(100, 200)$ . It may also be noted that  $z$  is maximum at each point of the line segment joining them. Thus the problem has an infinite number of (finite) optimum solutions. This happens when the objective function is “parallel” to one of the constraint equations.

## *Exceptional Cases*

Maximize  $z = 5x_1 + 7x_2$

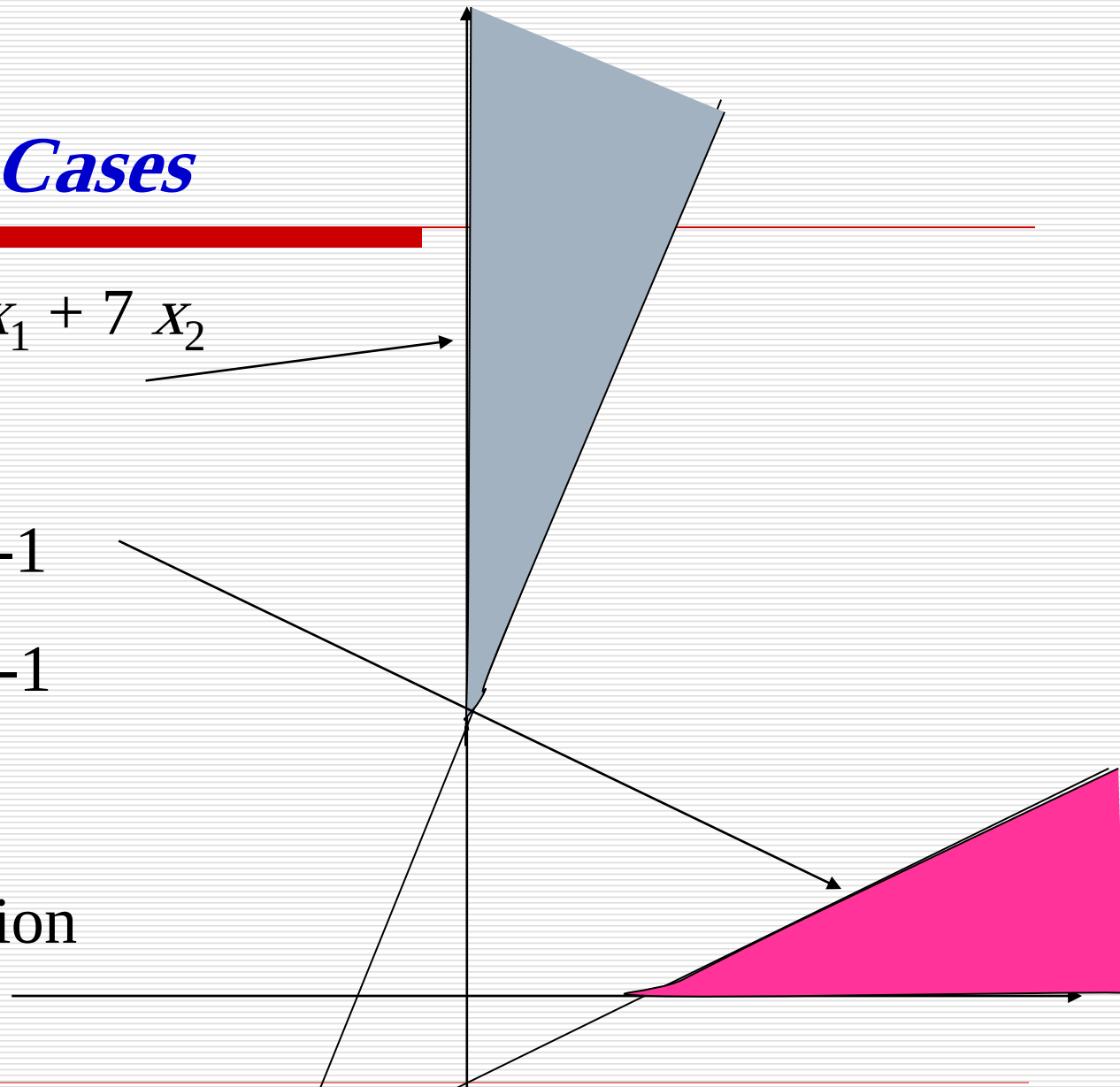
Subject to

$$2x_1 - x_2 \leq -1$$

$$-x_1 + 2x_2 \leq -1$$

$$x_1, x_2 \geq 0$$

No feasible solution



## ***unbounded solution***

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Maximize  $z = x_1 + x_2$

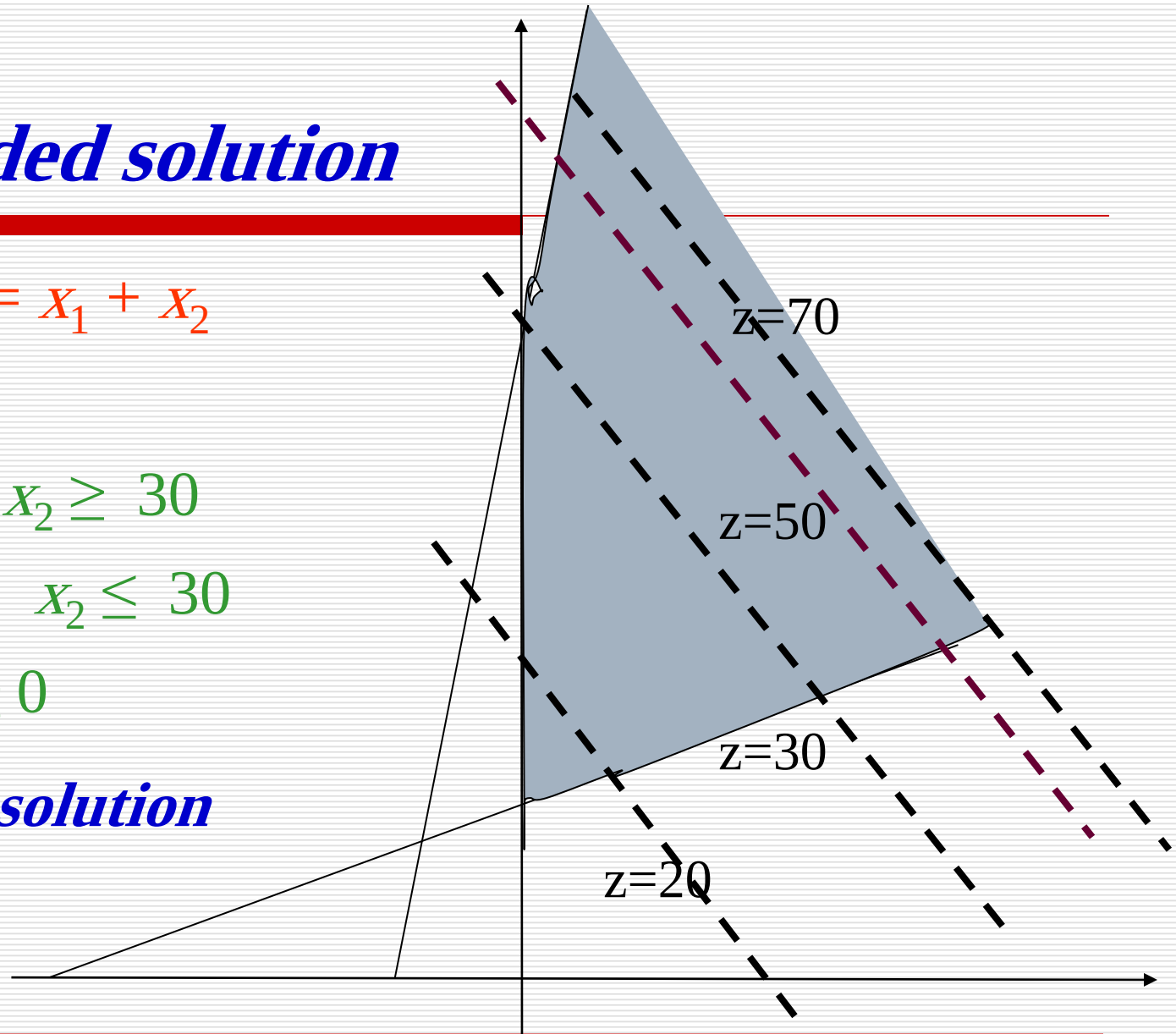
Subject to

$$-x_1 + 3x_2 \geq 30$$

$$-3x_1 + x_2 \leq 30$$

$$x_1, x_2 \geq 0$$

***unbounded solution***



## TORA Moment.

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- ❑ The menu-driven TORA graphical LP module should prove helpful in reinforcing your understanding of how the LP constraints are graphed.
- ❑ Select Linear Programming from the MAIN menu. After inputting the model, select Solve => Graphical from the SOLVE/MODIFY menu.
- ❑ In the output screen, you will be able to experiment interactively with graphing the constraints one at a time, so you can see how each constraint affects the solution space.